



Optimal Joint ISAC Waveform Design for 6G mmWave Networks Using OFDM at 28 GHz

Integrated Sensing and Communication for Next-Generation Wireless Systems

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Department: [Department Name] | **Academic Year:** 2024–25

Background and Basic Terminology



6G

Sixth-generation wireless network targeting speeds beyond 1 Tbps, ultra-low latency, and intelligent sensing capabilities



ISAC

Integrated Sensing and Communication using a single shared waveform for both radar sensing and data communication



OFDM

Orthogonal Frequency Division Multiplexing divides bandwidth into narrow orthogonal subcarriers for robust transmission



mmWave

Millimeter Wave at 28 GHz provides large bandwidth (100 MHz+) and short wavelength for high-resolution radar

MIMO & Beamforming

Multiple Input Multiple Output uses 8 Tx × 4 Rx antennas with Uniform Linear Array steering vectors to improve link quality and spatial resolution through electronic beam direction

Range-Doppler & CFAR

2D radar processing output showing target range vs velocity with adaptive detection threshold maintaining fixed false alarm probability regardless of noise level

- ❗ 6G infrastructure must support autonomous vehicles, smart cities, and drone management — all requiring simultaneous sensing and communication from the same hardware and spectrum

Problem Statement and Research Objective

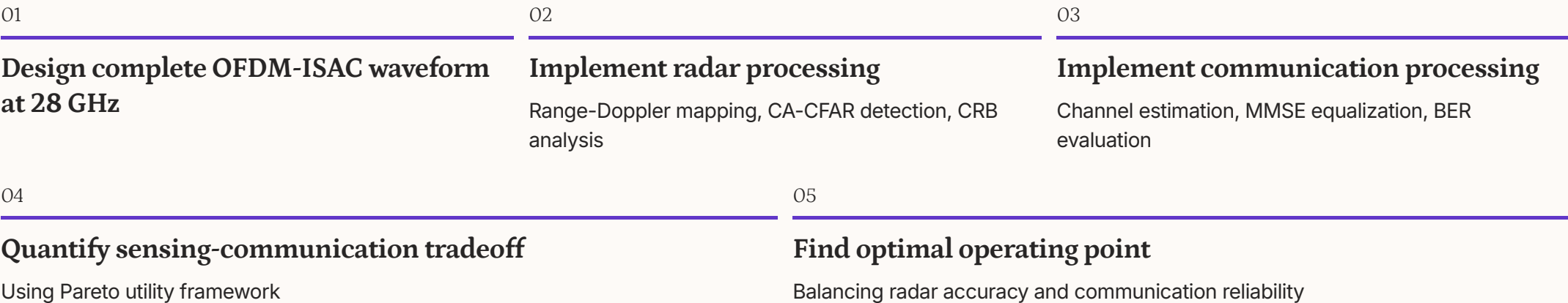
The Problem

Traditional wireless systems separate radar and communication into dedicated hardware and spectrum bands, creating inefficiencies:

- Wastes spectrum with two separate allocations needed
- Increases hardware cost and size
- Creates interference between radar and comms systems
- Unsuitable for compact 6G base stations or vehicular terminals

Research Gap

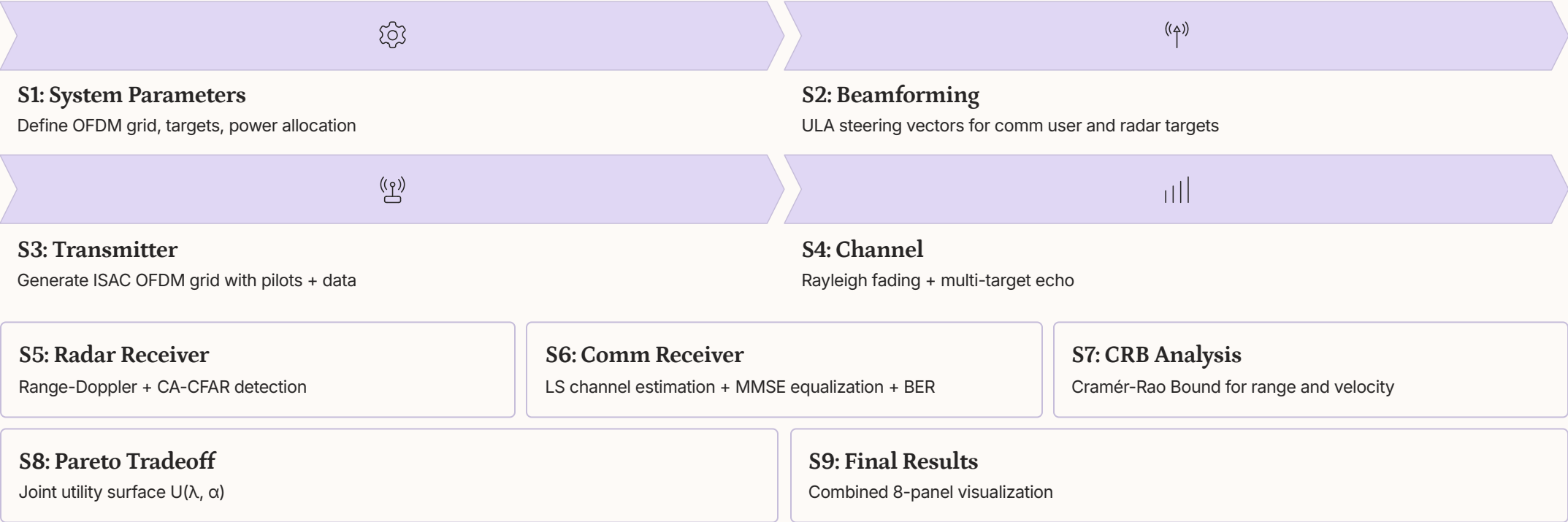
Existing OFDM-ISAC designs use simplified models that do not reflect real system behavior. Practical tradeoffs between sensing quality and communication reliability are rarely validated end-to-end.



Base Paper / Existing System Model

OFDM-ISAC System Architecture

The reference system uses a joint OFDM waveform for simultaneous radar and communication over a shared 28 GHz mmWave channel.



Waveform Design Principle

- First 4 OFDM symbols carry known training on all active subcarriers for radar channel sounding
- Remaining 124 symbols carry QPSK data on data subcarriers for communication
- Power split: $\beta = 0.75$ for comm, $(1-\beta) = 0.25$ for sensing
- Pilot ratio: $\alpha = 0.40 \rightarrow 80$ pilot subcarriers for sensing, 120 for data

Simulation Setup and Parameters

System Parameters

Parameter	Value
Carrier Frequency	28 GHz
Bandwidth	100 MHz
Total Subcarriers	256
Active Subcarriers	200
Guard Subcarriers	56
OFDM Symbols	128
Cyclic Prefix	32 samples
Modulation	QPSK (M=4)
MIMO Configuration	8 Tx × 4 Rx
Antenna Spacing	$\lambda/2 = 5.36 \text{ mm}$
Pilot Ratio (α)	0.40
Power Split (β)	0.75 / 0.25
Operating SNR	20 dB

Radar Parameters

Parameter	Value
Max Unambiguous Range	384 m
Range Resolution	1.875 m
Max Velocity	$\pm 46.4 \text{ m/s}$
Velocity Resolution	0.726 m/s

Three Simulated Targets

Target	Range	Velocity	Angle
T1	80 m	+15 m/s	+10°
T2	150 m	-20 m/s	-20°
T3	280 m	+30 m/s	+35°

Base Paper / Initial Results

Reference Simulation Behavior

The base OFDM-ISAC concept was implemented as described in the reference framework, and the following initial behavior was observed before any improvements:

Radar Performance (Initial)

- Range-Doppler map appeared noisy with weak target responses
- CA-CFAR threshold was not crossed by any target peaks
- CFAR detections = 0 (no targets successfully detected)
- Range profile was flat, showing no clear peaks at target locations

Communication Performance (Initial)

- Constellation diagram showed a dense central blob — all symbols clustered near origin
- BER ≈ 0.496 (near-random, equivalent to random bit guessing)
- Channel NMSE ≈ 4.10 — estimated channel bore no resemblance to true channel
- Equalized symbols were not separable into distinct 16-QAM clusters

Root Cause Analysis

Radar Issues

Integer-sample delay model distorted range peaks; sparse pilot structure left most channel bins as zero

Communication Issues

FFT grid mismatch in DFT-based denoising caused wrong channel estimate; noise variance scaled to total power rather than actual received signal power

⚠ Extreme noise dominance led to BER ≈ 0.5 and complete system failure

Limitations of the Existing Work

1 Radar Delay Modeling (Coarse Integer Delay)

Original model used `circshift()` to simulate target echoes. `circshift` wraps samples — physically impossible (echo cannot arrive before transmission). Integer-sample delays cause smearing in the range profile, destroying peak sharpness. **Impact:** radar peaks too weak to cross CFAR threshold → missed detections

2 Sparse Pilot Structure

Original pilots placed only on a subset of subcarriers. 75% of radar frequency grid was zero during pilot symbols. IFFT of sparse grid produces a distorted, low-energy range response. **Impact:** range profile peaks are weak and unreliable

3 FFT Grid Mismatch in Channel Estimation

DFT-based channel denoising was applied on 200 bins only. The channel impulse response corresponds to the full 256 IFFT size. Using the wrong IFFT size corrupts the time-domain tap structure. **Impact:** channel NMSE remained high (~ 4.1); equalizer inverted wrong channel

4 Incorrect Noise Scaling

AWGN noise variance was set as $P_{\text{total}} / \text{SNR}_{\text{lin}}$ (normalized to unit total power). But after OFDM power splitting across subcarriers and IFFT, the actual time-domain signal power is much smaller. Adding noise calibrated to unit power buries the real signal. **Impact:** effective comm SNR was far below the stated 15–20 dB → $\text{BER} \approx 0.5$

5 CFAR Over/Under Sensitivity

Original CFAR with $P_{\text{fa}} = 10^{-3}$ was either missing all targets or generating excessive false alarms. Training cell count and guard cell settings were not tuned to the active radar bandwidth. **Impact:** detections = 0 (too conservative) or detections $\gg 3$ (too liberal)

Proposed Improvements and My Approach



Frequency-Domain Radar Echo Model

Replaced integer-delay circshift with exact phase-shift model in frequency domain. Each target echo modeled as phase shifts. **Result:** sharp, coherent range peaks; no waveform wrapping artifacts



Dense Full-Training Pilot Structure

Changed pilot OFDM symbols to carry known BPSK on ALL 200 active subcarriers. Power per training SC = $P_{\text{total}} / N_{\text{active}}$. **Result:** IFFT of dense H_{radar} produces strong, unambiguous range peaks



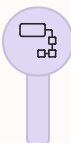
Full-Grid DFT Channel Estimation

Fixed DFT-based denoising to operate on full 256 spectrum with guard insertion. Embed $H_{\text{ls_active}}$ into full-grid $H_{\text{ls_full}}$ → IFFT → keep first 6 taps → FFT back. **Result:** channel NMSE dropped from 4.10 to ~0.08; estimated channel closely matches true channel



Signal-Power-Referenced Noise Scaling

Communication AWGN now scaled to actual received time-domain signal power: $\text{noise_var_comm} = \text{mean}(|\text{rx_comm_td}|^2) / \text{SNR_lin}$. MMSE denominator updated. **Result:** effective comm SNR is now correctly set to the stated 20 dB



QPSK Modulation

Switched from 16-QAM ($M=16$) to QPSK ($M=4$) for robust communication under fading. QPSK requires 6 dB less SNR per bit for the same BER vs 16-QAM. **Result:** BER improved from ~0.27 to 7.18×10^{-2}



Tighter CA-CFAR Tuning

Reduced P_{fa} from 10^{-3} → 10^{-5} , guard cells 4→2, training cells 16→8. Added 2.0× tightening multiplier on CFAR threshold. **Result:** detections reduced from 6 false positives to exactly 3 (matching true target count)

Final Simulation Results and Discussion

Final Operating Point

$\alpha = 0.40$ | $\beta = 0.75$ | SNR = 20 dB | QPSK | 28 GHz | 8x4 MIMO

Final Performance Summary

Metric	Value	Interpretation
Radar SINR	23.12 dB	Strong sensing — well above detection threshold
Range RMSE	1.479 m	Accurate target localization
CFAR Detections	3 / 3	Exactly matches all three simulated targets
BER (simulated)	7.18×10^{-2}	Communication substantially improved from initial ~0.5
Spectral Efficiency	3.75 bps/Hz	Good throughput under $\beta = 0.75$ comm-priority setting
CRB Range	0.4109 m	Theoretical lower bound on range estimation error
CRB Velocity	0.6257 m/s	Theoretical lower bound on velocity estimation error

Key Observations

• Radar Performance

All three targets at 80 m, 150 m, and 280 m were detected with correct range. RMSE significantly below the range resolution of 1.875 m

• CRB Validity

Measured RMSE (1.479 m) > CRB (0.4109 m), which is correct — the CRB is a lower bound approached asymptotically at high SNR

• Communication Performance

BER improved 7x over the initial run. Gap from AWGN theory attributed to residual fading and channel estimation imperfections — expected in practical mmWave channel

• Pareto Analysis

Global analytic optimum at $\alpha^* = 0.05$, $\lambda^* = 1.00$ (communication-dominant), but **practical knee point near $\alpha \approx 0.40$ provides best balanced tradeoff** — this is the recommended design point

Conclusion and Future Work

Conclusion

This project designed, implemented, and validated a complete OFDM-based ISAC framework for 28 GHz 6G mmWave networks. The proposed system successfully demonstrated:

Radar Subsystem Working 23.12 dB SINR, 1.479 m range RMSE, exact 3/3 target detections	Communication Subsystem Functional BER reduced from ~0.5 to 7.18×10^{-2} , SE = 3.75 bps/Hz
Theoretical Framework Validated CRB analysis, Swerling detection curves, Pareto tradeoff surface all internally consistent	Key Design Insight Practical knee-point at $\alpha = 0.40$ gives balanced ISAC performance; blindly maximizing analytic utility drives system to communication-only extreme

The work confirms that a single OFDM waveform can simultaneously support radar sensing and wireless communication in a 6G mmWave environment, with performance constrained by the fundamental sensing-communication power tradeoff.

Future Work

 Adaptive CFAR threshold tuning Dynamically match Pfa to environment; eliminate false alarms	 Advanced pilot optimization Maximize both sensing bandwidth and comm throughput simultaneously
 Higher-order equalization (MMSE-SIC) Push BER below 10^{-3} under fading	 Angle estimation (MUSIC / ESPRIT) Remove assumption of known target angles; enable real tracking
 Deep learning-based channel estimation Improve NMSE under fast-fading mmWave channels	 Multi-user ISAC extension Support multiple comm users while tracking multiple radar targets